

The Power of The Web and The Internet

Web vs. Internet

The terms “*Web*” and “*Internet*” may seem identical, either by definition or purpose. Technically, they describe different concepts, but a better comprehension of these ideas and their history leads to new realizations of their functionalities.

Web (World Wide Web)

It is a network of web pages that are found using a link address. It is called the “*web*” based on its interconnected nature, designed to easily get various resources.

The web allows access to an array of documents that are interconnected through hypertext or hypermedia links, i.e., hyperlinks and electronic connections that link similar pieces of information, allowing the user to smoothly transition from one page to another.

Brief History of the Web

Tim Burners-Lee

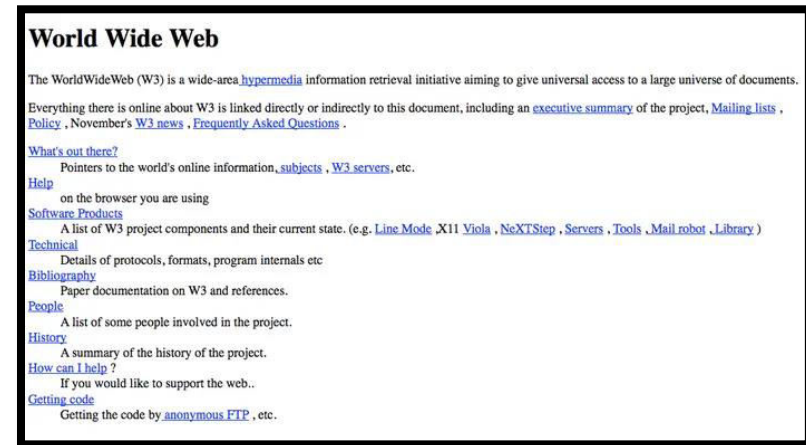
He is a British scientist who invented the Web, also called World Wide Web (WWW), in 1989 while working at CERN. The fundamental idea was to combine the growing technologies of computers, data networks, and hypertext into a powerful and simple global information system use.

He wrote the first proposal for the Web in March 1989 and the second in May 1990. By November 1990, it was formalized as a management proposal with the help of Robert Cailliau, a Belgian systems engineer.

By the end of 1990, he had the first Web server and browser running at CERN while developing the code for his Web server on a NeXT computer. To prevent it from being accidentally switched off, he put up a hand-written label in red ink that says “*This machine is a server. DO NOT POWER IT DOWN!!!*”

The First-Ever Website

The first web page went live on August 6, 1991, addressed as <http://info.cern.ch/hypertext/WWW/TheProject.html>. It contained links to the WWW project itself, including a description of a hypertext, technical information on the web server creation, and links to other web servers once they become available.



Source: <https://www.businessinsider.com/flashback-this-is-what-the-first-website-ever-looked-like-2011-6>

Figure 1. The closest copy of the first-ever website

Extension and Going Global

In March 1991, the WWW software became available to colleagues that use CERN computers. By August 1991, Burners-Lee announced his work on Internet newsgroups, making the interest in the project spread across the world.

It was evident that the small team at CERN could not handle all the work, so he launched a plea for developers to join. By 1993, the National Center for Supercomputing Applications (NCSA) released versions for PC and Macintosh environments. The existence of reliable and user-friendly browsers on these emerging computers had an immediate impact on the phenomena of the world wide web.

The Internet

It is a massive interconnection of networks connecting millions of computers globally, thus forming a network wherein any computer can communicate with other computers connected to it.

It is also a network of physical cables including copper telephone wires, TV cables, and fiber optic cables. Wireless connections such as Wi-Fi and 3G/4G are considered networks as they rely on these physical cables to access the Internet.

The process of internet connection starts with entering a website. The computer then sends a request over the wires to a server where websites are stored, and once the request arrives, the server recovers the website and sends the retrieved data back to the computer.

Brief History of the Internet

The 1960s

In 1962, computer scientist Joseph Carl Robnett Licklider, started an idea for a global computer network. Works by Leonard Kleinrock, Thomas Merrill, and Lawrence G. Roberts on packet-switching theory pioneered the world's first wide-area computer network. Roberts then published a plan for Advanced Research Projects Agency Network (**ARPANET**), making it a standard for all military computer networking.

The 1980s

In 1982, the PhoneNet system was founded which is both connected to the ARPANET and the first commercial network, Telenet. It broadened internet access allowing email communication between various nations in the world.

Paul Mockapetris, Jon Postel, and Craig Partridge developed the **Domain Name System** to control the increasing number of users on the internet using domain names. By 1984, a computer manufacturer registered *symbolics.com*, becoming the first domain.

The 1990s

ARPANET was decommissioned in the 1990s. Tim Berners-Lee and his colleagues at CERN develop **hypertext markup language** (HTML) and the **uniform resource locator** (URL), giving birth to the first version of the World Wide Web.

In 1995, Microsoft launched Windows 95. Online businesses like Amazon, Yahoo, and eBay followed suit, and Java is created, enabling animation on websites and starting a new wave of internet activity.

The 2000s

The 2000s sees the rise and the burst of the *dot-com bubble* or the Internet bubble. It is the period between 1995-2000 when investors financially supported Internet-based startups hoping they would all turn massive profits. By 2001, most *dot-com* companies were gone.

The 2000s also saw Google's meteoric rise to domination in the search engine market, besting Yahoo!, and MSN Search.

This decade saw the emergence of Wi-Fi, a wireless Internet communication, as well as mobile internet devices like smartphones.

Web vs. Internet

Simply put, the web is an application built on the internet. The internet is the infrastructure that connects computer networks, while the web is a system of information accessible through the Internet.

Not everything available on the internet requires the web as there are platforms that do not rely on it, such as email. Sending an email is possible, regardless of the web or a web browser, as it uses SMTP (Simple Mail Transfer Protocol) which is a secure email relay service.

How the Web and the Internet Changed the World

The impact of the web lies in making internet-connected devices widely accessible, paving the way for the current digital age. This accessibility made the world's population into internet users.

Communication

After the benefits of telephones and televisions, one of the biggest changes in communication was caused by the web and the internet. Messages sent via mail can take days up to weeks to arrive, but with the use of email and social media sites, easier and instant communication has become available.

Unavoidably, this also created an environment where careless communication is present. It is instant as it is destructive. Thoughts can be communicated quicker, making misunderstanding a risk through impulsive messaging. It is advisable to re-think messages concerning the other person before hitting the send button.

Information

Traditional books still exist, but so are the problems with accessibility. The web and the internet gave the world access to various information resources, from old books to newly published books in an e-book format.

Access to general information is also made possible through search engine sites like Google; allowing research and learning of new skills and knowledge that was once limited, due to cost and inaccessibility.

However, the downside of this is the false information, or "fake news" being spread with barely any fact-checking. Misinformation aims to deceive and create an alternative truth by twisting facts for someone else's favor. It is encouraged to fact-check everything on the internet using reputable websites and trusted resources.

Entertainment

Far from the limited disk and cassette players and expensive channels in cable networks, the entertainment industry has been one of the most impacted areas of the web and the internet.

Aside from the entertainment brought by social media applications such as Facebook and Twitter, the emergence of gaming has risen rapidly, as seen on gaming websites such as Twitch and YouTube Gaming. Newspapers and magazines are now digitalized, providing an easier and cheaper copy of the physical ones.

Streaming services such as Netflix, Disney+, Spotify, and Apple Music have served as the main source of entertainment for the past years, which is far from the limited media from before.

The adverse effect is the absence and lack of real human connection and interaction. Online entertainment consumption continues to increase, creating an unhealthy habit that leads to fatigue. It is encouraged to only consume entertainment for an observed amount of time and spend more time outside of that bubble.

Education

Web and the Internet played a significant role in lockdowns caused by the COVID-19 pandemic, which forced every education systems to adapt to an online learning model. Video conferencing tools were effective for online classes, while online learning platforms such as the NEO eLMS portal were utilized to assess student skills remotely.

One of the effects it created reflects on how students became too reliant and too comfortable in their classes without applying the same enthusiasm during regular classes. This can be avoided through focused learning and parent and instructor supervision.

References:

- Britannica (2019). *World wide web*. [Web Article]. Retrieved on July 11, 2022, from <https://www.britannica.com/topic/World-Wide-Web>
- CERN (n.d.a.). *A short history of the Web*. [Web Article]. Retrieved on July 22, 2022, from <https://home.cern/science/computing/birth-web/short-history-web>
- Google Arts & Culture (2022). *The world wide web: The invention that connected the world*. Retrieved on July 11, 2021, from <https://artsandculture.google.com/story/the-world-wide-web-the-invention-that-connected-the-world/>
- Jefferson Online (2017). *From ARPANET to the world wide web: An internet history timeline*. [Web Article]. Retrieved on July 11, 2022, from <https://online.jefferson.edu/business/internet-history-timeline/>